DS5006 Machine Learning — Report

Data Perturbations: How it Affects Satisfaction Prediction

Sub-Question 1 (SQ1) Evaluation: Numerical Meaning of Ordinal Rating Scales

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1 Experimental Setup

Context & Objective: In the given airlines satisfaction data, service ratings qualitative ordinal items (0–5).i.e. integers from 1-5$(0, 1, 2, 3, 4, 5)$, assuming equal interval spacing. This component addresses **Scientific Question 1 (SQ1)**: *Does the numerical meaning of an ordinal rating scale matter?*

Hypothesis & Rationale: Satisfaction predictions will not be invariant to numerical spacing, diverging across model paradigms. Tree-based partitioning methods rely on univariate order statistics and split thresholds $\{x_j \leq \theta\}$, invariant to strictly monotonic transformations. In contrast, linear and kernel methods evaluate inner products $\mathbf{w}^\top \mathbf{x} + b$, where numerical spacing directly alters decision margins, gradient dynamics, and regularization.

Benchmark Dataset: Kaggle Binary Aviation Satisfaction dataset ($N = 699{,}635$, 22 columns; target `satisfaction`: 55.6% dissatisfied, 44.4% satisfied). Predictors include exactly **13 ordinal service ratings** (12 features on a 0–5 scale, where 0 denotes unrated/N/A, whereas *Baggage handling* has no 0 values and is strictly on a 1–5 scale), 4 continuous variables (*Age*, *Flight Distance*, *Departure/Arrival Delays*), and 4 categorical variables (*Gender*, *Customer Type*, *Travel Type*, *Class*).

Preprocessing Pipeline: Implemented in an isolated `ColumnTransformer`: (i) Median imputation and standard scaling ($\mu = 0, \sigma = 1$) for linear models, or passed unscaled to tree models; (ii) Most-frequent imputation and one-hot encoding for categorical variables.

Controlled Monotonic Manipulations: Evaluated 4 strictly order-preserving mappings $g(\cdot)$: (1) **Original (Linear):** $\{0, 1, 2, 3, 4, 5\}$ (Baggage: $\{1..5\}$); (2) **Mild (Convex):** $\{0, 1, 2.2, 3.6, 5.2, 7\}$ (Baggage: $\{1, 2, 3.2, 4.6, 6.2\}$); (3) **Strong (Superlinear):** $\{0, 1, 2, 4, 7, 11\}$ (Baggage: $\{1, 2, 3, 5, 8\}$); (4) **Exponential (2^k):** $\{0, 1, 2, 4, 8, 16\}$ (Baggage: $\{1, 2, 3, 5, 9\}$). Monotonicity was strictly verified over all 699,635 rows: $g(v_k) < g(v_{k+1})$. Baggage handling uses a dedicated 5-point mapping because it contains no zero/N/A ratings.

Models & Validation Protocol: Evaluated 5 models over 5 repeated stratified splits (seeds $\{42, 123, 2025, 7, 99\}$, test $N = 139{,}927$, train $N = 559{,}708$): (i) **Majority Baseline** (empirical mode); (ii) **Logistic Regression** (L2, $C = 1.0$, L-BFGS, standardized); (iii) **Linear SVM** (`LinearSVC`, $C = 1.0$, 2-fold Platt sigmoid calibrated); (iv) **Decision Tree** (CART, max depth 15, min leaf 20); and (v) **Random Forest** (100 trees, max depth 20, min leaf 10, $\sqrt{p}$ features). Identical splits and seeds were reused across encodings to enable paired passenger-level tracking.

2 Experimental Recordings

A total of 100 model runs (5 splits × 4 encodings × 5 models) and 13,992,700 out-of-fold inferences were logged. Table 1 reports the exact recordings, and Figure 1 plots sensitivity dynamics.

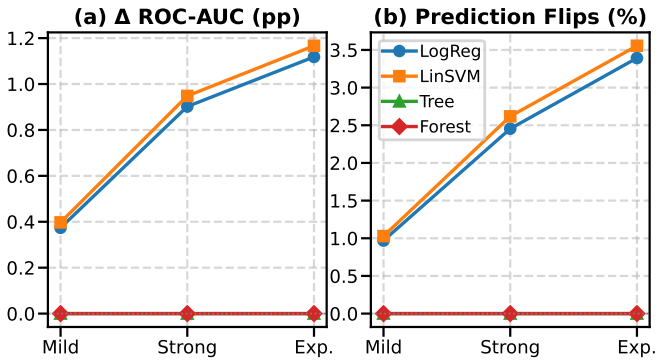


Figure 1: Sensitivity Dynamics Across Encodings. (a) Paired ROC-AUC gain (ΔROC-AUC in percentage points). (b) Test prediction flip rate (% of passenger classifications changed). Tree architectures remain flat at 0; linear models show marked gains and prediction volatility.

Table 1: Experimental Results Across Encodings. Mean metric across 5 stratified splits ($N_{\text{test}} = 139{,}927$ per split; std ≤ 0.0012). Paired metrics (Δ, $|\Delta p|$, Flips) are evaluated on identical test passengers.

Model	Metric	Orig.	Mild	Str.	Exp.	Max Flips
Logistic Regr.	ROC-AUC	0.9216	0.9254	0.9307	**0.9328**	—
	Bal. Acc.	0.8756	0.8807	0.8884	**0.8918**	**3.39%**
	Mean $\lvert\Delta p\rvert$	—	1.23%	3.19%	4.67%	(4,746)
Linear SVM	ROC-AUC	0.9207	0.9247	0.9302	**0.9324**	—
	Bal. Acc.	0.8738	0.8792	0.8877	**0.8912**	**3.56%**
	Mean $\lvert\Delta p\rvert$	—	1.27%	3.30%	4.75%	(4,974)
Decision Tree	ROC-AUC	0.9525	0.9525	0.9525	0.9525	—
	Bal. Acc.	0.9188	0.9188	0.9188	0.9188	**0.00%**
	Mean $\lvert\Delta p\rvert$	—	0.00%	0.00%	0.00%	(0)
Random Forest	ROC-AUC	**0.9570**	**0.9570**	**0.9570**	**0.9570**	—
	Bal. Acc.	**0.9225**	**0.9225**	**0.9225**	**0.9225**	$\leq 10^{-4}$%
	Mean $\lvert\Delta p\rvert$	—	0.00%	0.00%	0.00%	(≤ 1)
Majority	ROC-AUC / BA	0.5000	0.5000	0.5000	0.5000	0.00%

- **Decision Tree:** ΔROC-AUC = **0.000000** and ΔBalanced Accuracy = **0.000000** across all encodings. Exactly 0 test predictions flipped (0.00%).
- **Random Forest:** ΔROC-AUC $\leq 10^{-6}$ and ΔBA $\leq 10^{-6}$ across all splits. Prediction flip rate $\leq$ 0.0001% ($\leq$ 1 passenger out of 139,927, due to floating-point split tie-breaking).
- **Logistic Regression:** ROC-AUC rises monotonically: 0.92164 → 0.92538 → 0.93066 → **0.93282** (+1.12 pp gain). Balanced Accuracy rises 0.87558 → **0.89176** (+1.62 pp). Flipped 3.39% of test passengers (4,746 per split) with 4.67% mean probability drift.
- **Linear SVM:** Parallels LogReg: ROC-AUC rises 0.92074 → **0.93240** (+1.17 pp gain) and Balanced Accuracy rises 0.87377 → **0.89121** (+1.74 pp). Flipped 3.56% (4,974 test passengers per split) with 4.75% mean drift.